ZS-A29

Dust and Vacuum from Measure/Form Actions

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A Cross-Carrier No-Go, the Rank-83 Flux Obstruction, the Three Gates of the 83/121 Bridge, and the Convergence of Independent Algebraic and Dynamical Routes on a Present-Epoch Budget

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Dependencies: ZS-A28 v2.0 (B3-D; import-and-relabel discipline; Bousso-Polchinski [8]), ZS-A26 v2.2 (everpresent target; three risks), ZS-A27 (A-Q-Only No-Go), ZS-A23 (dimension-weighted semigroup; rank-to-energy embedding OPEN), ZS-A20, ZS-A19, ZS-M19 §10 (X/Y PK, HYPOTHESIS-strong), ZS-A7/F14/M3 (topological 2pi/4pi), ZS-M1/M12 (z*, lambda, Koenigs, damped spiral - sub-Planck), ZS-F10/U8 (tau_n=tP exp(n pi/A)), ZS-F9 (rho_Z=0), ZS-U4 (w=-1 attractor), ZS-F2, ZS-F23.
Repository: github.com/KennyKang-git/zspin

Verification Summary

Verification: 56 fail-closed asserts PASS (*verify_zs_a29_v1_6.py*), fully fail-closed, counts printed by the script and quoted verbatim: **30 load-bearing**, **23 new mechanism** (four-form K1-K6; supertrace L1-L4; rank-to-energy M1-M2; i-tetration orbit T1-T3; emergence tests E1-E4; and the P3 *coexistence* result CX1-CX4, Appendix E.5), and **3 bookkeeping/consistency** (the 4.235 round-trip; F3, now reclassified as an algebraic-consistency check), plus 4 printed notes that are not asserts. **Two v1.2 verification nits fixed:** v1.2's K6 was a tautology (bool(w+1==w+1)); v1.3 replaces it with a *real* Levi-Civita computation ($\mathrm{eps}_{(m\,a\,b\,g)}\mathrm{eps}_{n}^{(a\,b\,g)}$ = -3! g_{mn} -> T_{mn} = -$\mathrm{rho}_{L}\mathbf{g}_{mn}$ -> w = -1); and F3 (an algebraic-consistency check) is moved out of load-bearing. (**A**, **Q**, dim **Z**) = (35/437, 11, 2) LOCKED.
The honest skeleton (this version's organizing result): the bridge rank $\mathbf{P}_{L}$ = 83 => Omega_{L} = 83/121 is *not* one calculation but **three independent gates** - G1 projector *selection* (TFS), G2 flux *collectivization* (the rank-83 flux has up to 83 free constants, §2.3), G3 rank-to-energy *normalization* (§8.2) - plus the single-parent field *identification* (§2.4) and PT. v1.0 saw “one calculation left”; v1.2-v1.3 show a multi-gate structural problem. The flagship 83/121 is *not* advanced.
What v1.6 consolidates (the manuscript is now integrated, per prior AI-pass feedback). (1) Title corrected (v1.2's “Single-Parent” contradicted its own §2.4 “not yet single-parent”). (2) K6 made real; F3 reclassified. (3) §2.3 “Bousso-Polchinski landscape” softened to a *kinematic precursor* (the multi-flux freedom exists prior to flux quantization and membrane dynamics). (4) §4.1 sign argument corrected (the positive-semidefinite Maxwell action + $\mathbf{P}_{L}$ selection fix the sign, *not* the rank trace). (5) The “only exit is

external" over-statement is balanced by listing the **internally-decidable tasks** (§8.3), including a computed rank-to-energy candidate (the maximally-mixed state, M1-M2). (6) The user's **i-tetration orbit reading of w(z)** is computed in full (Outlook §12, O5): the damped spiral is real, but under the corpus' own time anchor it gives a *frozen* w ~ -1, **not** the DESI crossing - so it reinforces *w* = -1, and the zero-parameter DESI claim does not follow. Heavy process/meta material is consolidated into the Appendices.

§0. Abstract

ZS-A28 reduced the vacuum-identification bridge **B3-D** to an IMPORTED-PROVEN field-theory half (a covariant top-form on the rank-83 complement $\mathbf{P}_L$ forces *w* = -1, constant rho_L) and an identification half whose load-bearing antecedent is the present-epoch coincidence (UN). A29 is an honest audit of that gate; it advances correctness and structure, not the flagship number 83/121.

Construction and No-Go (§2-§3). The Guendelman-Nissimov-Pacheva measure yields dust and a dynamical rho_{DE} = 2M from one action. Separating spacetime form-degree (4) from internal rank (83), the vacuum carrier is realized as a projector-valued four-form; on a real 121x121 rank-83 projector its quadratic form gives $rho_L = (1/2)|\mathbf{P}_L\mathbf{M}|^2 >= 0$ with the rank-38 flux dropping out (PROVEN-as-construction: the realization that the GNP measure *can* be rank-83 projector-valued is ASSUMED, F-A29.1; *w* = -1 is IMPORTED). The **cross-carrier No-Go** (Theorem A29.2) stands and is *stronger*: not only is rho_{DE} = 2M a free integration constant, the entire rank-83 flux vector is free (a Bousso-Polchinski-*type* multi-flux freedom, prior to quantization).

The three gates (§8). The organizing result of this version: rank $\mathbf{P}_L$ = 83 does **not** by itself give $Omega_L$ = 83/121. It requires **G1** projector selection (why $\mathbf{P}_L$, not $\mathbf{P}_m$), **G2** flux collectivization (reduce 83 free constants to one), and **G3** rank-to-energy normalization (why the energy fraction equals the rank fraction). A computed candidate for G3 is the **maximally-mixed state** $rho_* = \mathbf{I}/121$, for which $Tr(\mathbf{P}_L rho_*) = 83/121$ exactly - reducing G3 to the decidable, *internal* question "is the cosmic vacuum the de Sitter-thermal maximally-mixed state on M_{121}?" So external review is *necessary* but not the *only* route; §8.3 lists three internally-decidable tasks. **Scale and dynamics (§4-§6).** The everpresent / geometric-mean / de Sitter forms are one Friedmann relation written four ways (PROVEN-algebraic; CKN is IMPORTED-MOTIVATED, not a scale derivation). The present-epoch selection is an undecided fork; DESI remains a NON-CLAIM.

The i-tetration orbit reading (§12, O5; computed). A new direction: read the de Sitter attractor *w* = -1 as the i-tetration **fixed point** z* (n -> infinity), matter (rank-38) as the **orbit** still flowing in, and the free constant M as "where we are on the orbit" = the observable scale factor. The multiplier is lambda = (i pi/2)z*, |lambda| = 0.8915 < 1 (a *damped* spiral; Koenigs verified, T1). This resolves the ZS-U4 tension by making *w* = -1 the asymptotic fixed point. **But the load-bearing test fails:** under the corpus' own anchor $tau_n = t_P \exp(n\, pi/\mathbf{A})$, only 0.4% of one spiral oscillation advances across the DESI redshift range (z ~ 0.5 -> 0), so *w* is **frozen** ~ -1 there (T3) - the spiral reinforces *w* = -1, it does **not** produce DESI's phantom-past evolution, and the "zero-parameter DESI w(z)" claim does not follow (the n <-> scale-factor map is the free, UNDERIVED part). Status: HYPOTHESIS-strong for the reframing; the DESI prediction is **not** supported by computation. Net: B3-D unconditional closure PARTIAL; 83/121 unchanged. (**A**, **Q**, dim **Z**) LOCKED.

The dynamical route and the convergence (this version). Dropping top-down number-fitting, we ask whether the budget *emerges* from the i-tetration rule alone. It does not, instructively. A bare i-tetration lattice is a pure contraction (it erases all matter); the minimal rule that restores dust/vacuum coexistence is a bistable source whose threshold is *corpus-tied*, $J > J_c = 4(1 - |$**lambda**$|) = 0.434$ (Maxwell point $J_M = (9/2)(1 - |$**lambda**$|)$), but the dust/vacuum *ratio* is NOT fixed - one phase generically wins, and even Maxwell balance is unstable to curvature-driven coarsening. So the emergence route reaches the **same conclusion** as the algebraic No-Go, from the opposite direction: across **four independent routes** (A27 and A29.2 No-Go theorems, the rank-83 flux obstruction, and reaction-diffusion coexistence) the budget (6,32,83)/121 is a **present-epoch boundary condition, not a dynamical invariant**. The one structure that *would* fix the ratio is a **rank-weighted master equation** whose probability-conserving stationary state is (38/121, 83/121) with no free parameter - the dynamical form of the maximally-mixed candidate - but it stays HYPOTHESIS-strong (it assumes equal per-channel amplitudes and an occupation-to-energy map). (**A**, **Q**, dim **Z**) LOCKED.

Epistemic Status Legend

PROVEN - established by internal computation/algebra at machine precision.
PROVEN-as-construction - a construction is exhibited and its algebra verified, but a nontrivial realization claim it depends on is assumed.
PROVEN-algebraic-identity / CONSISTENCY - an exact algebraic rewriting or a self-consistency check; not a derivation of new physics.
IMPORTED-PROVEN / IMPORTED-MOTIVATED - proven externally and reused / supplies only motivation or heuristic, not a derivation.
DERIVED-CONDITIONAL - DERIVED given explicitly named conditions (selection / saturation / sign convention).
HYPOTHESIS-strong / -weak - pre-registered and corpus-non-conflicting, with a named undatured assumption / a missing bridge or pending anti-numerology check.
TARGET / COMPUTED-INCOMPLETE - a decidable computation set up but deferred, or begun but not reproducing the target.
OPEN (gate) - not closed by current tools; a sharply stated, internally-decidable obstruction.
CLOSED-NEGATIVE / NO-GO - a route proven not to work.
NON-CLAIM / WITHDRAWN - explicitly not asserted, or a prior statement removed by this audit.

§1. Introduction

Z-Spin Cosmology fixes cosmology's dimensionless ratios from **A** = 35/437 and **Q** = 11 = (2, 3, 6). The absolute vacuum scale B3 splits into B3-A (A-Q-only, CLOSED-NEGATIVE), B3-B (absolute value, OPEN/terminal), B3-C (today's relation), and B3-D (the bridge). A28 handed A29 the present-epoch coincidence (gate XB-6b). This paper's positive content is a sharp negative result (the cross-carrier No-Go, §3) and a structural map of what closing 83/121 would require (the three gates, §8). Its central honest finding, across versions, is that the No-Go - not the value of any coefficient - is the load-bearing obstruction: the present vacuum normalization is a free integration constant (indeed a free rank-83 flux vector), so geometry fixes the *ratio structure* but not the *scale*. Process and audit history are recorded in Appendices C-D; this body is kept to the physics.

§2. The measure/form action and the projector-valued top-form (Theorem A29.1, as construction)

§2.1 The GNP measure

From S_{GNP} = integral sqrt(-g)L_1 + integral Phi(B)L_2 [1-3], a pressureless dust and a dynamical rho_{DE} = 2M (M the integration constant of the four-form B_3).

§2.2 Form-degree vs internal rank, and honest status

Form-degree-4 (spacetime) does not populate an internal rank-83 subspace (algebra). Realize the vacuum carrier as a projector-valued four-form S_F = -(1/2*4!) integral sqrt(-g) $\mathbf{F}^A$ $(\mathbf{P}_L)_{AB}\mathbf{F}^B$. On a real 121x121 rank-83 $\mathbf{P}_L$ (K1-K2), rho_L = (1/2)$|\mathbf{P}_L\mathbf{M}|^2$ = (1/2) $sum_{a=1..83}$ M_a^2 >= 0, with all 38 $\mathbf{P}_m$-fluxes dropping out (K3-K4). **PROVEN-as-construction**: the positivity/confinement is projector algebra; the nontrivial claim - that the GNP measure four-form *can* be the rank-83 projector-valued object - is **ASSUMED** (F-A29.1). w = -1 is **IMPORTED** (Henneaux-Teitelboim [4]; verified at the tensor level in K6: $eps_{(m\,a\,b\,g)}eps_n^{(a\,b\,g)}$ = -3! g_{mn} forces $\mathbf{T}_{mn}$ = -$rho_L\mathbf{g}_{mn}$). TFS is DERIVED-CONDITIONAL on the projector *selection* (contracting $\mathbf{P}_m$ is equally valid).

§2.3 The rank-83 flux obstruction (kinematic precursor of a multi-flux freedom)

With A = 1, ..., 121 and rank $\mathbf{P}_L$ = 83, the on-shell flux $\mathbf{P}_L\mathbf{M}$ has **83 independent components** and rho_L = (1/2) $sum_{a=1..83}$ M_a^2 (K5). **Stated carefully (softened from v1.2):** this is the *kinematic precursor* of a Bousso-Polchinski-*type* landscape [8] - the unconstrained projector-valued construction has up to 83 free flux components - *prior to* the flux quantization, distinct charges, and membrane dynamics that a full BP landscape adds. Even at this kinematic level the conclusion holds: the rank fraction 83/121 does **not** fix the energy normalization; a **flux-collectivization** structure (O(83) isotropy, a single collective mode $\mathbf{P}_L\mathbf{M}$ = $m_L\mathbf{v}_L$, or a rank-to-energy theorem) is required. This **strengthens** the No-Go (§3) and is gate G2 of §8. **[OPEN]**

§2.4 The action is not yet single-parent

S_{GNP}'s four-form B_3 and the projector-valued A_3^A are, as written, **two** fields. If two, the model has two cosmological terms; if one, the GNP measure must be promoted to the algebra-valued object and its hidden symmetry / dust current re-verified. A genuinely single-parent action S_{grav} + S_{GNP}[g, phi, $\mathbf{B}$] + $S_{constraint}$[$\mathbf{P}_m$, $\mathbf{P}_L$, $\mathbf{B}$] (one algebra-valued $\mathbf{B}_3$ generating both) must be written and varied in g, phi, B. v1.3 does not yet write it (the title is corrected accordingly). **[OPEN - this is why the title says “Measure/Form Actions,” not “Single-Parent.”]**

§3. The cross-carrier No-Go (Theorem A29.2)

Theorem A29.2 (CLOSED-NEGATIVE). rho_{matter}/rho_L = (38/83)*(2 $mu_Z n/\mathbf{ZF}f^2$); the bracket scales under mu_Z -> alpha mu_Z but changes under f -> beta f (I1-I3); rho_{DE} = 2M is an integration constant (I4). With §2.3, the vacuum normalization is free as the whole rank-83 flux vector (K5).
Unified-Normalization is **not** a theorem of the action; it is a present-epoch condition. This is the one

unambiguously load-bearing, closed result of A29 (as a No-Go). It is also why “zero free parameters”, defensible for the dimensionless *ratios*, does *not* extend to the absolute *scale*. (QED)

§4. The everpresent scale is Friedmann-class

PROVEN-algebraic. With c^2 = Omega_L = 83/121, $\text{rho}_L = 3c^2\mathbf{M\text{-}bar}_P^2 H_0^2$ = (geometric-mean)^4 = 24 $\text{pi}^2\mathbf{M\text{-}bar}_P^4/S_{dS}$ (F1-F2, E3). Both the CKN-saturated bound (L = 1/H) and the de Sitter entropy *reduce to this same Friedmann relation* (F3, now classed as a consistency check), so neither *independently sets* the scale; the scale *is* Friedmann-at-the-horizon, near-tautological once Omega_L ~ O(1). CKN is **IMPORTED-MOTIVATED** (a collapse heuristic for Omega_L <= O(1)). The only Z-Spin-specific content is the coefficient $\text{chi}_Z/\text{alpha}_{patch} = (3*83/121)^2$ = 4.235 **[COMPUTED-INCOMPLETE]**; its round-trip (G1-G2) is bookkeeping, not a computation of chi_Z.

§4.1 What fixes the sign (corrected from v1.2)

Corrected. v1.2 wrote that the positive rank trace 83/121 > 0 fixes the sign. That is wrong - *every* projector has a positive rank, and it does not say why the *vacuum* projector rather than the matter projector carries the energy. The sign is fixed instead by the **positive-semidefinite Maxwell-type four-form action** together with $\mathbf{P}_L$ being a projector ($\text{rho}_L = (1/2)|\mathbf{P}_L\mathbf{M}|^2 >= 0$) *once*$\mathbf{P}_L$ is selected and a ghost-free energy convention is adopted. The rank trace fixes the channel *multiplicity*, not the sign. Status: **DERIVED-CONDITIONAL** on the positive Maxwell sign and the $\mathbf{P}_L$ selection (G1).

§5. The Physical-Trace gate (DERIVED-CONDITIONAL)

Case A (121, q_L = 0.6860) and Case B (120, 0.6833) both lie within ~1% of Omega_L ~ 0.6847 (B1-B4); data does not discriminate. Establishing the ZS-F23 seam mode is BRST-closed-not-exact needs the $H^0(Q_{BRST})$ cohomology [11]. Status: DERIVED-CONDITIONAL argument; the cohomology is a TARGET.

§6. The present-epoch selection as an undecided fork

Branch A (constant Lambda, frozen w = -1): the crossing epoch is an integration constant; coincidence OPEN. **Branch C (everpresent):** Omega_L ~ O(1) always; coincidence dissolves, at the cost of w != -1. **DESI: NON-CLAIM (kept).** DESI DR2's phantom-past $w_0 w_a$ fit is reproduced by neither branch; the everpresent fluctuation fails **independently in amplitude and in redshift-dependence**. (v1.1's “modulus/phase pair” description was withdrawn in v1.2 as a pattern-fit; a real function w(z)+1 has no canonical modulus/phase split.) $|w+1| \sim 10^{-61}$ is A26's, not re-derived here. The i-tetration reading of §12 (O5) is the corpus' best candidate for a *secular* w(z), but - computed - it gives a frozen w ~ -1, not the DESI crossing.

§7. The fermion/boson supertrace direction (HYPOTHESIS-weak; Outlook §12)

The X = fermion / Y = boson reading exists (ZS-M19 §10) but is HYPOTHESIS-strong and “reinterpreted, not re-derived”; the corpus' deeper f/b machinery is topological (2pi/4pi; NC-A7.8

disclaims SUSY); Pauli exclusion is not the CKN gravitational exclusion that sets the floor (§4). Using Coleman-Weinberg [12, 13], the three divergence gates are kept separate (Str $\mathbf{M}^0$ = Str 1 quartic; Str $\mathbf{M}^2$ quadratic; Str $\mathbf{M}^4$ log), with the **dimensional** form Str $\mathbf{M}^4/M_*^4 = 4.235$ (L4). The coincidence Str $\mathbf{M}^0 = 6 - 2*3 = 0$ (L2) would cancel the **M-bar**$_P^4$ quartic ($c_0 \to 0$), but §4 uses that same **M-bar**$_P^4$ as a live ceiling - whether the two compose or compete is **UNRESOLVED**, and the 6 = 2*3 dof assignment inserts a Weyl factor on sector dimensions (a numerology trap until derived). HYPOTHESIS-weak; collected in §12 (O3).

§8. The three gates of the 83/121 bridge, and the internally-decidable tasks

The most useful result of the v1.0-v1.3 audit is not a closure but a **decomposition**: what looked in v1.0 like "one calculation" (compute $\text{chi}_Z/\text{alpha}_{\text{patch}} = 4.235$) is, correctly, a chain of **independent gates**, each of which must be closed for rank $\mathbf{P}_L = 83$ to become the energy fraction $\text{Omega}_L = 83/121$.

§8.1 The gates

G1 - projector selection (TFS). Why does $\mathbf{P}_L$ (rank 83), not $\mathbf{P}_m$ (rank 38) or any other projector, carry the vacuum top-form? Not fixed by the four-form structure (contracting $\mathbf{P}_m$ is equally consistent). DERIVED-CONDITIONAL.
G2 - flux collectivization. The rank-83 flux has up to 83 free constants (§2.3, K5). Reducing them to a single normalization needs an O(83) isotropy, a single collective mode, or a rank-to-energy theorem. OPEN.
G3 - rank-to-energy normalization. Even granting G1, G2, why does the *energy* fraction equal the *rank* fraction (the present-epoch UN of A28)? Matter dilutes; the vacuum is constant; the equality holds only on one hypersurface. OPEN.
Beyond these three sit the single-parent field **identification** (§2.4) and **PT** (§5). So v1.0's single missing calculation is, accurately, **five gates**. Exhibiting this structure - rather than asserting the value - is the audit's principal positive contribution; it is what an external reader needs to attack the problem.

§8.2 A computed candidate for G3: the maximally-mixed state

ZS-A23 carries a dimension-weighted semigroup but leaves the physical embedding Omega_i = rank $\mathbf{P}_i/121$ OPEN. A concrete candidate (verified M1-M2): the **maximally-mixed state** $\text{rho}_* = \mathbf{I}_{121}/121$ on the 121-channel face algebra gives

$\text{Tr}(\mathbf{P}_L\,\text{rho}_*)$ = rank $\mathbf{P}_L/121 = 83/121$, $\text{Tr}(\mathbf{P}_m\,\text{rho}_*) = 38/121$.(A29.5)

So the rank-to-energy bridge is realized *exactly* by one specific state - the maximally-mixed (infinite-temperature) state. This is **not** a closure (the identity $\text{Tr}(\mathbf{P}_i\mathbf{I}/121)$ = rank/121 holds for *any* projector, so it does not by itself single out the vacuum). Its value is that it **reduces G3 to a sharp, internally-decidable question:** *is the cosmic vacuum the maximally-mixed (de Sitter-thermal) state on M_{121}?* The de Sitter horizon is thermal, which makes this physically natural and connects to the everpresent/holographic reading - but it must be *derived* from the parent-state dynamics, not assumed. HYPOTHESIS-strong.

§8.2b The rank-weighted master equation (the best G3 candidate)

The maximally-mixed candidate has a natural *dynamical* realization. If the transition rate into a sector is proportional to its number of microstates (its rank) - equivalently, if per-channel amplitudes are equal by the Z-Spin seam symmetry - then the parent occupations obey

d/dtau $(\mathbf{p}_m, \mathbf{p}_L)^T = k\,[-83, +38 ; +83, -38]\,(\mathbf{p}_m, \mathbf{p}_L)^T$, $q_{m\text{->}L} = 83k$, $q_{L\text{->}m} = 38k$.(A29.6)

Its unique, probability-conserving stationary state is $\mathbf{p}_m = 38/121$, $\mathbf{p}_L = 83/121$ (verified ME1), with *no free parameter* - exactly what the cubic source of §13/E.5 cannot do (there the ratio is free). It is the *dynamical* form of rho$_*$ = $\mathbf{I}$/121, unifying three threads: the maximally-mixed candidate, ZS-A23 dimension-weighted semigroup, and the inverse (slog-type) outward channel as the L->m rate. **Honest status: HYPOTHESIS-strong, not DERIVED.** Two gaps remain, both the same equal-weight assumption in different clothes: (g1) the rank-weighted rate = the vacuum being the maximally-mixed state is *assumed*, not proven; and (g2) $\mathbf{p}_L$ is an *occupation* - identifying it with the energy fraction Omega$_L$ needs a state-to-stress-energy map (equal energy per microstate), the G3 normalization gap itself. The master equation *relocates* G3 to a sharp dynamical question - *does the parent-state dynamics actually have equal-amplitude, rank-weighted rates?* - rather than closing it. [Best current candidate for G3; surfaced in an AI-assisted pass as the dynamical realization of the maximally-mixed candidate.]

§8.3 The internally-decidable tasks (external review is necessary, not the only route)

v1.2 over-stated that "the only exit is external." External specialist review **is** necessary for validation, but the internal programme retains **three decidable mathematical tasks** that do not require it: (i) **unified-action variation** - write $S_{\text{grav}} + S_{\text{GNP}} + S_{\text{constraint}}$ with one algebra-valued $\mathbf{B}_3$ and vary in g, phi, B (§2.4); (ii) **flux-collectivization analysis** - a group-representation study of whether an O(83) symmetry or single collective mode is forced (G2); (iii) **single-parent stationary-state embedding** - test whether the parent-state dynamics drive rho toward rho$_*$ = $\mathbf{I}$/121, closing G3 (§8.2). These are the highest-value next steps *alongside* external contact, not in place of it.

§9. The B3 classification after A29 v1.6

Table 9.1. Status after ZS-A29 v1.6. down = demotion, up = raise, NEW = surfaced; G1/G2/G3 = the three bridge gates of §8.

Item	Status (v1.6)	Change vs v1.2 / basis
B3-A (A-Q-only)	CLOSED-NEGATIVE	unchanged (A27)
B3-B (absolute scale)	OPEN - terminal	unchanged
Four-form mechanism	PROVEN-as-construction	unchanged; K6 now REAL (Levi-Civita) not a tautology
w=-1 of the top-form	IMPORTED-PROVEN	K6: eps eps=-3! g => T=-rho g => w=-1 (computed at tensor level)
G1 projector selection (TFS)	DERIVED-CONDITIONAL	gate named (§8.1)
G2 flux collectivization	OPEN (gate)	§2.3 softened: kinematic precursor of BP-type freedom (not full landscape)

G3 rank-to-energy	OPEN (gate); candidate	NEW (§8.2): max-mixed state rho*=I/121 gives 83/121 EXACTLY (M1-M2); reduces G3 to a decidable question
Single-parent identification	OPEN	§2.4; TITLE corrected to "Measure/Form Actions"
Everpresent scale	Friedmann-class / IMPORTED-MOTIVATED	unchanged from v1.2
Mean+variance: sign	DERIVED-CONDITIONAL	down from v1.2 "DERIVED": Maxwell sign + P_L selection fix it, NOT rank trace (§4.1)
PT (121 vs 120)	DERIVED-CONDITIONAL	unchanged (BRST = TARGET)
DESI / present-epoch fork	NON-CLAIM / OPEN	unchanged; i-tetration (O5) computed -> frozen w, not DESI
i-tetration orbit w(z) (O5)	HYPOTHESIS-strong	NEW (§12): spiral real (T1-T2); corpus anchor -> frozen w (T3); reinforces w=-1, NOT DESI
Fermion/boson supertrace	HYPOTHESIS-weak	unchanged; dimensional fix kept (§7)
chi_Z/alpha_patch = 4.235	COMPUTED-INCOMPLETE	unchanged; NOT advanced
P3 coexistence (emergence)	DERIVED-COND / OPEN	NEW (§13/E.5): minimal source J>J_c=4(1-|lambda|); ratio NOT fixed (curvature coarsening even at Maxwell) -> 4th route to the No-Go
Rank-weighted master eq (G3)	HYPOTHESIS-strong	NEW (§8.2b): stationary (38/121,83/121), no free J = dynamical rho*=I/121; gaps: equal-amplitude + occupation->energy
Budget (6,32,83)/121	present-epoch boundary condition	CONVERGENCE: 2 No-Go theorems + flux obstruction + coexistence all agree it is NOT a dynamical invariant
B3-D unconditional closure	PARTIAL	net: structure clarified (5 gates + master-eq candidate); number unmoved

§10. Falsification gates

Mathematical / immediate. (F-A29.1) If the GNP measure provably cannot be rank-83 projector-valued, §2.2 fails as a realization. (F-A29.2) If no single-parent action reproduces dust+vacuum without interference, §2.4 fails.
Structural / consistency. (F-A29.3) If no flux-collectivization structure exists (G2), 83/121 is not an energy prediction. (F-A29.3b) If parent-state dynamics do NOT drive rho -> **I**/121, the G3 candidate (§8.2) fails. (F-A29.4) If $\text{chi}_Z/\text{alpha}_{\text{patch}}$!= 4.235 on (3,2,6)/11, Branch C's promotion fails.
Observational / external. (F-A29.5) The i-tetration w(z) (O5) is falsifiable in principle by its zero-parameter *decay-per-oscillation* ratio (~0.85; T2), but this needs >= 2 resolved DE oscillations, beyond DESI DR2; and under the corpus anchor it predicts a *frozen w* ~ -1 (T3), so a robust DESI *evolution* (wa != 0) at high significance would disfavor the corpus' natural reading. (F-A29.6) $\text{dN}_{\text{eff}}^{\text{BBN}}$ = 2**A** = 0.160 and *r* = 0.0089 inherited; always-on $\text{dN}_{\text{eff}}^{\text{CMB}}$ = 0.160 already FALSIFIED. (F-A29.7) Omega_L outside [0.6833, 0.6860] +/- error kills both PT cases.

§11. Conclusion

A29 v1.6 consolidates a multi-version audit whose central, honest result is a **convergence**: four independent routes - two No-Go theorems (A27 A-Q-only; A29.2 cross-carrier), the rank-83 flux obstruction (K5), and now the dynamical reaction-diffusion coexistence analysis (§13/Appendix E.5) - all reach the same conclusion, that the budget (6,32,83)/121 is **not a dynamical or geometric invariant but a present-epoch boundary condition**. The supporting structural results: the **cross-carrier No-Go** (§3, strengthened to a free rank-83 flux vector); the **three-gate decomposition** of the 83/121 bridge (§8 - the audit's principal scientific contribution, turning v1.0's "one calculation" into a precise five-gate map); a **computed G3 candidate** (the maximally-mixed state, §8.2); and a **full computation of the i-tetration orbit reading** of *w*(z) (§12), which honestly *reinforces***w** = -1 (resolving the ZS-U4 tension by reading it as the spiral's fixed point) while showing - by computation, under the corpus' own anchor - that it does **not** produce the DESI crossing.
The flagship 83/121 and $chi_Z/alpha_{patch}$ = 4.235 are **unchanged**; the absolute scale is terminal in B3-B. Three rounds of audit have moved honesty, not the physics number - which is itself the finding: *the obstruction is the No-Go and the five gates, not a missing coefficient*. The next steps are now sharp and divided honestly: **three internally-decidable tasks** (unified-action variation, flux-collectivization analysis, stationary-state embedding; §8.3) that the author can pursue, *and* external domain-expert contact for the parts that internal iteration cannot settle (whether the GNP measure can be rank-83 projector-valued; whether a flux-collectivization theorem exists). (**A**, **Q**, dim **Z**) = (35/437, 11, 2) LOCKED.

§12. Outlook (HYPOTHESIS directions, explicitly NOT results)

(O1) Flux collectivization (G2); (O2) single-parent unification (§2.4); (O3) supertrace dof bridge (§7); (O4) a secular w(z) carrier. Each is HYPOTHESIS-weak and detailed above or in v1.2.

§12.1 (O5) The i-tetration fixed-point/orbit reading of w(z) - computed

The proposal. Read the de Sitter attractor *w* = -1 not as a separate top-form but as the i-tetration **fixed point** z* (the n -> infinity limit of the corpus' self-referential map i^z); read matter (rank-38) as the **orbit** still flowing toward it; and read the free integration constant M of §2-§3 as "where we are on the orbit" = the **observable scale factor**. This is attractive: it would convert a free parameter into an observable, and it reads ZS-U4's *w* = -1 as the asymptotic fixed point (resolving that fork). The corpus has z*, the multiplier lambda, Koenigs linearization, and the damped spiral - but only in the sub-Planck auto-surgery (ZS-M1/M12/F14); transplanting them to late-time *w*(z) is the new step.
The computation (this paper). The multiplier is lambda = (i pi/2) z* = -0.566 + 0.688 i, with **|lambda| = 0.8915 < 1** (so convergence is a *damped* spiral - Koenigs ratio verified, T1) and arg lambda = 129.4 deg (so *w*(z) approaches -1 as a **spiral**, crossing -1 periodically: a *deterministic, secular* phantom crossing, not a stochastic one). Two signatures are **mapping-invariant** (fixed by z* alone, zero free parameters, T2): one full oscillation is 2.78 iterations, and each successive excursion of *w* from -1 is ~0.85x the previous.
The honest verdict (why it does not yet deliver DESI). The load-bearing assumption is the map *iteration index n <-> cosmological scale factor / e-folds*, which is **UNDERIVED**. Testing the corpus' own candidate anchor $tau_n = t_P \exp(n\, pi/\mathbf{A})$ (ZS-F10/U8): it gives A/pi = 0.025 iterations per e-fold, today n_0 = 3.6, and across the DESI redshift range (z ~ 0.5 -> 0) only **0.4% of one oscillation** advances (T3). So under the corpus anchor the spiral is **frozen**: *w* ~ -1 = const over the observable range - it **reinforces***w* =

-1 (consistent with ZS-U4 and ~ΛCDM), and does **not** reproduce DESI's phantom-past evolution (which needs wa != 0). Producing the DESI crossing would require a *faster*, non-corpus n <-> scale map - i.e. new free input - so the "zero-parameter DESI w(z)" claim does **not** follow; the corpus anchor actively favors frozen w = -1. **Status: HYPOTHESIS-strong** for the reframing (fixed point = vacuum, orbit = matter, M = observable position, U4 tension resolved); the DESI-w(z) prediction is **not supported by computation**. The one clean falsifiable prediction - the 0.85 decay-per-oscillation ratio - needs >= 2 resolved DE oscillations, beyond DESI DR2. The decidable next step is to *derive* the n <-> scale-factor map (or prove it cannot be the slow tau_n one) before any data comparison.

§13. A methodological pivot: from top-down fitting to bottom-up emergence

The deepest anti-numerology move available is to stop trying to *fit* 83/121, 35/437, and Q = 11 from above (counting polyhedron faces, integer partitions) and instead ask whether they **emerge** from the one dynamical rule the corpus most deeply commits to - the i-tetration self-referential rotation - applied to a *blank slate*. Three concrete proposals make this testable: (P1) **braiding / TQFT** - read i^z as a braid-group action on the 2D Z-boundary and ask whether **A** is a topological invariant (Jones/Chern) or an FQHE filling factor; (P2) **tensor-network thermodynamics** - start from a random qubit graph, evolve under i-tetration, and ask whether it spontaneously breaks into the (2, 3, 6) blocks, making Q = 11 a *theorem* rather than an axiom; (P3) **quantum cellular automata** - a 2D quaternion-spin grid with a local i-tetration update, asking whether dust and vacuum separate at the ratio 83/121.
This is the right direction - it makes the framework falsifiable *from below*, not just from observation - and it is pre-registered in full (with protocols and success criteria) in Appendix E. **But the toys, run honestly, fail, and instructively.** (E1-E2) The i-tetration orbit's rotation number (arg lambda / 2pi = 0.3596) is *not* a low-order rational, so the braid is quasi-periodic and **does not close** - no finite Jones polynomial exists for the actual orbit; and while 5/19 is a (high-flux) Jain state, 7/23 is not standard and a *product* of filling factors is not a filling factor, so the FQHE/Chern reading of **A** is unsupported (a Chern number must be an integer). (E3) A bare i-tetration coupled-map lattice, being a pure contraction (|lambda| < 1), collapses to a *single* attractor - dust and vacuum do **not** even coexist, let alone at 83/121; the split is either absent or a tunable artifact of the coupling. (E4) i^z is a map on C = R^2 (the Z-sector) with a single complex multiplier; it has no internal (2, 3, 6) spectrum, so it cannot generate the X and Y dimensions *alone*.
The honest lesson (and the value). None of the three numbers emerges from i-tetration without tuned input, and the failures are *mechanistic*, not accidental: a single 2D contraction cannot, by itself, produce a multi-sector dimensional split, a closed braid, or a robust two-phase coexistence. A genuine bottom-up theory therefore needs an **explicit additional rule** - one that (i) couples entanglement structure to dimension (for P2), (ii) supplies a source or a second basin so matter can resist the de Sitter collapse (for P3), and (iii) selects a periodic sub-orbit if any knot invariant is to exist (for P1). Specifying *that* rule is the real open problem; Appendix E states it precisely as the next pre-registered target. The perspectives' value is exactly this: they convert "we keep circling the answer" into a set of **decidable, falsifiable emergence tests** - and they have already ruled out the naive versions. **[Status: the pivot is DERIVED (a falsification-first reframing); each emergence claim is PRE-REGISTERED-TEST / OPEN; the naive realizations are NON-CLAIM, ruled out by E1-E4.]**

Acknowledgements and Code Availability

verify_zs_a29_v1_6.py is fully fail-closed (every entry a real computation; the v1.5 E4 identity moved to a note, E1 corrected, CX4 split, the rank-weighted master equation ME1 added) and *prints its own counts* (30 load-bearing + 23 new + 3 bookkeeping = 56 asserts, + 8 notes), quoted verbatim above. **All “review” / “adversarial pass” herein is AI-assisted (Claude / ChatGPT / Gemini) inside the author's workflow; no external or human domain-expert peer review has taken place.** This work used Anthropic Claude for the audit, the i-tetration computation, and drafting; the author assumes full responsibility for all content, including errors caught only on later passes.

Appendix A. Verification ledger (56 fail-closed asserts; counts printed by the script)

All entries are real computations in verify_zs_a29_v1_6.py: 30 load-bearing, 23 new mechanism, 3 bookkeeping, + 8 printed notes (not asserts).

Block	Check	Status
A-E (LB, 22)	rank budget; PT cases; A=(5/19)(7/23); N=69.16; 4N=276.64; rhoL/Mp^4=24pi^2/S_dS=7e-121	PROVEN
I1-I4 (LB, 4)	No-Go: rho_m/rho_L bracket scales under muZ, CHANGES under f; rho_DE=2M integration constant	PROVEN-symbolic
J1-J2 (LB, 2)	z*=0.4383+0.3606i; |f'(z*)|=0.8915<1	PROVEN
K1-K5 (NEW)	real 121x121 rank-83 P_L; rho_L=1/2 sum_{1..83}M_a^2; 38 P_m fluxes drop; 83 free flux constants (G2)	PROVEN-as-construction / OPEN
K6 (NEW, fixed)	REAL Levi-Civita: eps_(m a b g)eps_n^(a b g)=-3! g_mn => T_uv=-rho g_uv => w=-1 (was a tautology in v1.2)	IMPORTED, now tensor-verified
L1-L4 (NEW)	supertrace: no simple combo=4.235; Str M^0=Str 1 quartic (6-2*3=0); 3 gates; DIM fix Str M^4/M_*^4	HYPOTHESIS-weak
M1-M2 (NEW)	rho*=I/121: Tr(P_L rho*)=83/121, Tr(P_m rho*)=38/121 (rank-to-energy G3 candidate; max-mixed state)	HYPOTHESIS-strong
T1-T3 (NEW)	i-tetration: lambda=(i pi/2)z*, |lambda|=0.8915 spiral; 2.78 iter/osc & 0.85 decay (invariant); tau_n anchor => 0.4% of an osc over DESI range => FROZEN w	HYPOTHESIS-strong
E1-E3 (NEW)	emergence: NO low-period closure up to denom bound (NOT a no-Jones proof, E1 corrected); 5/19 Jain but 7/23 not => FQHE unsupported; bare i-tetration CML collapses to one attractor. (E4 = i^z is 2D: moved to a NOTE, was an identity)	PRE-REG TEST / NON-CLAIM
CX1-CX3,CX4a (NEW)	P3 coexistence: robust bistability J>J_c=4(1-|lambda|)=0.434 (J=J_c saddle-node); Maxwell J_M=4.5(1-|lambda|); phi+=2/3 generic (NOT 83/121); CX4a=analytic area sign (CX4b front-velocity = a NOTE)	DERIVED-COND / OPEN
ME1 (NEW)	rank-weighted master eq: stationary (38/121,83/121), conserves probability, no free J = dynamical rho*=I/121 (best G3 candidate); gaps g1 equal-amplitude + g2 occupation->energy (notes)	HYPOTHESIS-strong
G1-G2, F3 (BK, 3)	4.235 round-trip (circular); F3 reclassified algebraic-consistency (Hinf2 from rhoL)	bookkeeping

Notes (4)	printed commentary, NOT asserts (CKN reduces to Friedmann; no external review; flux landscape; G3 question)	NOTE

Appendix B. Cross-version safety and dependency check

A29 v1.3 uses (never modifies): ZS-A28, ZS-A26, ZS-A27, ZS-A23 (dimension-weighted semigroup; the §8.2 rho* candidate is a proposed closure of its OPEN embedding), ZS-A20, ZS-A19, ZS-M19 §10, ZS-A7/F14/M3, ZS-M1/M12 (z*, lambda, Koenigs - the §12 O5 computation transplants these from sub-Planck to late-time, a NEW use flagged HYPOTHESIS-strong), ZS-F10/U8 (tau_n anchor, tested in O5), ZS-F9, ZS-U4 (frozen w = -1 - O5 reads it as the asymptotic fixed point, a reinterpretation, not a modification), ZS-F2, ZS-F23. **Tensions recorded:** (i) Branch C / O5-secular would predict w != -1 (fork with U4, not adopted); but O5 computed under the corpus anchor gives frozen w = -1, so it is *consistent* with U4, not in tension. (ii) the §4/§7 **M-bar**$_P^4$ ceiling-vs-cancellation tension is flagged UNRESOLVED. **Relative to v1.2:** title corrected; K6 made real; F3 reclassified; §2.3 softened; §4.1 sign corrected; §8 three-gate structure + rho* candidate added; O5 computed. No downstream value changes. (**A**, **Q**, dim **Z**) LOCKED.

Appendix C. Internal adversarial-pass record (5-step, v1.3)

Step 0 - Long list (7). (0.1) the i-tetration orbit w(z) proposal - compute it fully; (0.2) title vs §2.4 contradiction; (0.3) K6 tautology + F3 classification; (0.4) §2.3 “BP landscape” too strong; (0.5) §4.1 sign argument; (0.6) “only external exit” over-statement + rank-to-energy candidate; (0.7) meta-audit density. **Dropped:** none substantive; (0.7) handled by consolidating meta into Appendices.
Step 1 - Issue list (4 MECE). (A) compute the i-tetration orbit and report honestly [0.1] - the one direction that could move a number; (B) editorial/verification corrections [0.2, 0.3, 0.4, 0.5]; (C) the gate structure + internal-task balance + rho* [0.6]; (D) meta density [0.7].
Step 2 - Issue tree. (A) is the lead (new physics, falsifiable); (B) are the fully-closable editorial fixes; (C) is the scientific reframing (three gates); (D) is presentation.
Step 3 - Traversal with status. (A) i-tetration: spiral PROVEN-real (T1-T2); DESI claim NOT supported (T3, frozen w); reframing HYPOTHESIS-strong. (B) title fixed; K6 real; F3 reclassified; §2.3 softened; §4.1 corrected. (C) three gates named (§8.1); rho* G3 candidate computed (§8.2, HYPOTHESIS-strong); internal tasks listed (§8.3). (D) meta -> Appendices.
Step 4 - Convergence. First pass N_changed ~ 9 (i-tetration computed, title, K6, F3, §2.3, §4.1, §8 gates, rho*, meta-trim). Second pass N_changed ~ 0. **CONVERGED**. The decisive new content (i-tetration) *self-limited*: computation showed it reinforces w = -1 rather than delivering DESI, so it did not inflate any claim - the anti-numerology discipline operated as designed.
Step 5 - Scoring. Convergent; corpus-non-conflicting; anti-numerology respected (the i-tetration DESI claim was tested and *rejected* by computation, not asserted). Leadership: the i-tetration computation was run before being asked to defend it, and reported against its own initial appeal - the honest form of leading. Honesty ~9.5; the i-tetration reframing is a genuine conceptual gain (M -> observable; U4 resolved); the physics number is still unmoved, now for a clearly-mapped five-gate reason.

Appendix D. Disposition of prior AI-assisted adversarial passes

Table D.1. Points raised by prior AI-assisted passes (Claude/ChatGPT/Gemini; NOT external or human peer review) and their disposition in v1.3.

Point (prior AI pass)	Disposition	Where
Title "Single-Parent" contradicts §2.4 "not yet single-parent"	ACCEPTED + FIXED	Title -> "Measure/Form Actions..."
v1.2 K6 is a tautology bool(w+1==w+1); F3 is algebraic-consistency not load-bearing	ACCEPTED + FIXED	K6 -> real Levi-Civita tensor; F3 -> bookkeeping (counts 30/15/3)
"precisely a Bousso-Polchinski landscape" too strong	ACCEPTED + SOFTENED	§2.3: kinematic precursor; prior to quantization/membranes
"positive rank trace fixes the sign" is wrong	ACCEPTED + CORRECTED	§4.1: Maxwell sign + P_L selection; status -> DERIVED-CONDITIONAL
"only exit is external" too strong; internal decidable tasks remain (rho*=I/121)	ACCEPTED + ADDED	§8.2-8.3: rho* G3 candidate (M1-M2) + three internal tasks
meta-audit content excessive for a physics paper	ACCEPTED	§1 trimmed; process consolidated into Appendices C-D
i-tetration spiral w(z) proposal (user breakthrough)	COMPUTED + INTEGRATED HONESTLY	§12 O5: spiral real (T1-T2) but frozen w under corpus anchor (T3); reframing HYPOTHESIS-strong; NOT a result
the loop has converged; physics numbers unmoved in 3 rounds; only real exit is external contact	ACCEPTED	§8.3, §11: internal tasks named AND external contact stated outstanding

Appendix E. Pre-registered bottom-up emergence tests (protocols, success criteria, toy results)

Each test below is stated so that it can be run by an independent party and can *fail*. "Toy result" is the outcome of a small computation in *explore_emergence.py*; "Pre-registered success" is the criterion fixed *before* any run, to prevent post-hoc fitting.

E.1 (P1) Braiding / topological invariant

Protocol. Treat the i-tetration orbit $z_{n+1} = i^{z\_n}$ near z* as a worldline braid on the 2D Z-boundary; compute the rotation number rho = arg(lambda)/2pi and, if rho is rational p/q, close the q-strand braid and evaluate its Jones polynomial / the associated Chern number; test whether **A** = 35/437 appears as that invariant or as an FQHE filling factor. **Pre-registered success:** A is reproduced as an integer Chern number or a standard (Jain/hierarchy) filling factor with no free choices. **Toy result (FALSIFIED):** rho = 0.3596 shows NO low-period closure up to the pre-registered denominator bound (nearest rational with denominator <= 12 is 4/11, residual 4e-3; <= 25 is 9/25, residual 4e-4) - which does NOT prove irrationality or that no Jones polynomial exists, only that no short braid closure was found; 5/19 is Jain (2p = 4) but 7/23 is not standard and the product of two filling factors is not a filling factor; a Chern number must be an integer. The FQHE/Chern reading is NON-CLAIM (E1-E2).

E.2 (P2) Tensor-network emergence of Q = 11

Protocol. Initialize a random qubit graph (no assumed (2, 3, 6) structure); evolve the entanglement under an i-tetration update; track the entanglement-entropy spectrum and any spontaneous block decomposition. **Pre-registered success:** the stable state breaks into exactly three sectors of effective dimension 2, 3, 6 (total Q = 11) with no dimension inserted by hand. **Toy / structural result (NOT available as stated):** i^z is a single map on C = R^2 with one complex multiplier - it carries no internal (2, 3, 6) spectrum, so the Z-dynamics cannot generate the X and Y dimensions by itself (E4). The test is only meaningful once an *additional* rule coupling entanglement structure to emergent dimension is specified; that rule is the open target.

E.3 (P3) Quantum cellular automaton: dust/vacuum separation

Protocol. A 2D grid of quaternion/Z_4 spin states {1, i, -1, -i} with a local i-tetration update; run from random initial data; classify cells as "vacuum" (settled at z*) vs "dust" (orbiting) and measure the steady-state fraction. **Pre-registered success:** the vacuum fraction converges robustly (independent of coupling/threshold) to 83/121 = 0.686, dust to 38/121. **Toy result (FALSIFIED):** a bare i-tetration coupled-map lattice is a pure contraction (|lambda| < 1) and collapses to a *single* attractor (~100% vacuum in 120 steps) - dust and vacuum do not coexist; where a partial split can be forced it tracks the arbitrary coupling, so 83/121 is a tunable artifact, not an invariant (E3).

E.4 What a genuine emergence theory must add

The three failures share one cause - a single 2D contraction is too simple to generate multi-sector structure - and so define the next pre-registered target precisely: an explicit rule that (i) for P2 couples entanglement to emergent dimension; (ii) for P3 supplies a **source or second basin** so matter (the orbit) resists the de Sitter collapse to z* (Appendix E.5 computes the minimal such rule: a bistable source of strength J >= 4(1-|lambda|); without it the contraction that gives *w* = -1 also erases all dust); (iii) for P1 selects a periodic sub-orbit so a knot invariant can exist. Until such a rule is specified and shown to yield the numbers with **no** tuned input, bottom-up emergence of (83/121, 35/437, Q = 11) is **OPEN / NON-DEMONSTRATED** - but it is now a set of sharp, falsifiable tests rather than an aspiration, and the naive versions are closed.

E.5 (P3, extended) The minimal source term and the conditions for coexistence

v1.4 found that the bare i-tetration lattice collapses to vacuum (no dust survives). v1.5 acts on the mechanistic diagnosis - matter needs to *resist* the de Sitter contraction - and adds the simplest such rule, then asks not whether 83/121 appears but what FORM of rule makes coexistence possible at all. Let phi in [0, 1] be a matter field (phi = 0 vacuum at z*, phi > 0 orbiting dust). The contraction rate is fixed by the corpus: gamma = 1 - |lambda| = 0.108. The minimal dynamics adds a bistable autocatalytic source to that contraction,

R(phi) = -gamma phi + J phi^2(1 - phi),   gamma = 1 - |**lambda**| = 0.108 (the i-tetration contraction rate).(E.1)

Computed result (explore_coexistence.py; CX1-CX4). (i) A second (dust) phase exists - and coexistence becomes possible - *iff* J >= $\mathbf{J_c}$ = 4(1 - |**lambda**|) = 0.434 (CX1-CX2); J = J_c is itself only a saddle-node creation threshold (phi+ = phi- = 1/2, non-hyperbolic), so robust bistability needs J > J_c. Below it only vacuum is stable, exactly reproducing the bare-lattice collapse. So, **within the minimal scalar cubic bistable ansatz**, the source threshold is fixed by the Z-Spin contraction rate (not a uniqueness theorem - other bistable, nonlocal, or conserved-field rules exist): a source whose strength exceeds a *corpus-tied* threshold set by the i-tetration contraction rate. (ii) The planar vacuum/dust front is stationary only at the Maxwell point $\mathbf{J}_M$ = (9/2)(1 - |**lambda**|) = 0.488 (CX3; analytic area-sign verified in CX4a, with the numerical front-velocity sweep in the explore script, CX4b): below it vacuum invades (dust -> 0), above it dust invades (dust -> 1). (iii) **The dust/vacuum ratio is NOT pinned:** generically it flows to 0 or 1; and at J_M only the *planar* front is stationary - a curved dust domain still SHRINKS by curvature-driven (Allen-Cahn) coarsening (verified: a disk loses ~40% of its area at J_M), so even Maxwell balance does not preserve a seeded fraction. A specific value such as 83/121 occurs only by *tuning* J and *seeding* the fraction - never as an output, and not even stably then. (The dust-phase field value phi_+ = 2/3 at J_M is a generic algebraic feature of the cubic, *not* the 83/121 volume fraction - an anti-numerology caution.)

The pre-registered J = f(A, Q) test (negative). A candidate counts as a derivation only if J is fixed by **A**, **Q** with no free choice AND the random-initial-condition ratio becomes 83/121 as an output. Neither holds: no simple **A**, **Q** expression selects J_c or J_M (these are fixed by |**lambda**| alone, but picking J = J_M is itself a selection), and even fixing J = J_M while *seeding* the ratio at 83/121 does not preserve it (it coarsens to ~1). So deriving J is **not** an exit. The structure that *would* fix the ratio is instead the conserved-probability **rank-weighted master equation** of §8.2b - which is why the resolution, if any, lies in parent-state dynamics, not in tuning a source strength.

Interpretation. This is a genuine advance on P3 - it converts “needs a source (open)” into a precise, falsifiable, corpus-tied condition for coexistence - and it carries an honest lesson: even with coexistence achieved, the ratio remains free. That *independently reproduces* the corpus' own Cross-Carrier No-Go (§3) and the G3 gate (§8): the dust/vacuum ratio is a **present-epoch condition, not a dynamical invariant**. The emergence route therefore meets the same obstruction as the algebraic route, from the opposite direction. **The next pre-registered target** is now sharp: derive the source strength J (or an additional selection principle pinning the front at a specific fraction) from **A** and **Q** - not a free knob. Until then, coexistence is **DERIVED-CONDITIONAL** (on J >= 4(1 - |**lambda**|)) and the ratio is **OPEN**.

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[ZS-A27] K. Kang, ZS-A27 v2.3: The Absolute Vacuum-Energy Scale and the A-Q-Only No-Go, Z-Spin Cosmology (2026).

[ZS-A23] K. Kang, ZS-A23: Dimension-Weighted Semigroup and the Rank-to-Energy Embedding (OPEN), Z-Spin Cosmology (2026).

[ZS-M1] K. Kang, ZS-M1: i-Tetration HSI Theorem, z*=0.43828+0.36059i, lambda=(i pi/2)z*, |lambda|=0.891514, Koenigs, Z-Spin Cosmology (2026).

[ZS-M12] K. Kang, ZS-M12: Auto-Surgery and the Damped Spiral (sub-Planck), Z-Spin Cosmology (2026).

[ZS-F10] K. Kang, ZS-F10: Information-Time Correspondence; tau_n = t_P exp(n pi/A), Z-Spin Cosmology (2026).

[ZS-M19] K. Kang, ZS-M19 v1.0: Database-PK reading of sector statistics (T8), Z-Spin Cosmology (2026).

[ZS-U4] K. Kang, ZS-U4: Locked late-time budget (6,32,83)/121, w=-1 attractor, Z-Spin Cosmology (2026).

Version History

v1.0-v1.1 (June 2026): GNP action + cross-carrier No-Go; v1.1 lowered five v1.0 overclaims and added the projector-valued four-form, but introduced a verification count mismatch and an “external peer review” mislabel.
v1.2 (June 2026): Fully repaired the verification (counts script-sourced); corrected the loop-honesty failure (disclaimer; “review” relabelled AI-assisted); lowered the four-form to PROVEN-as-construction and the everpresent scale to Friedmann-class; withdrew the modulus/phase description; surfaced the rank-83 flux and single-parent gates.
v1.3 (June 2026): Internal AI-assisted audit + i-tetration computation. Consolidated from internal Z-Spin Collaboration notes through v1.3.0. **Title corrected** (“Measure/Form Actions...”; v1.2's “Single-Parent” contradicted its own §2.4). **Verification**: K6's v1.2 tautology replaced by a real Levi-Civita tensor computation (eps eps = -3! g => w = -1); F3 reclassified as an algebraic-consistency check; counts 30 LB + 15 new + 3 bookkeeping = 48, script-printed. **Lowered/corrected**: §2.3 “Bousso-Polchinski landscape” -> kinematic precursor (prior to flux quantization/membranes); §4.1 sign argument -> the positive Maxwell action + $\mathbf{P}_L$ selection fix the sign (DERIVED-CONDITIONAL), not the rank trace. **Added (scientific)**: §8, the **three-gate decomposition** (G1 selection / G2 collectivization / G3 rank-to-energy) of the 83/121 bridge, with a computed G3 candidate - the maximally-mixed state rho$_*$ = $\mathbf{I}$/121 giving 83/121 exactly (M1-M2) - and three internally-decidable tasks (§8.3), balancing v1.2's

“only external exit” over-statement. **Computed (O5)**: the user's **i-tetration fixed-point/orbit reading** of *w*(z) - lambda = (i pi/2)z*, |lambda| = 0.8915 damped spiral (T1), invariant 2.78 iter/oscillation and 0.85 decay (T2). Under the corpus anchor tau_n only 0.4% of an oscillation advances over the DESI range (T3), so *w* is frozen ~ -1: the spiral *reinforcesw* = -1 (resolving the U4 tension as its fixed point) and does NOT yield the DESI crossing; the zero-parameter DESI claim does not follow. HYPOTHESIS-strong for the reframing; the DESI prediction is unsupported by computation. **Unchanged**: 83/121, chi_Z/$alpha_{patch}$ = 4.235 (COMPUTED-INCOMPLETE), B3-B terminal. Process material consolidated into Appendices C-D. No new fitted parameters. (**A** = 35/437, **Q** = 11, dim **Z** = 2) LOCKED.

v1.4 (June 2026): Methodological pivot + pre-registered emergence tests. Adds §13 and Appendix E: the three “blank-slate” proposals (P1 braiding/TQFT, P2 tensor-network Q = 11 emergence, P3 quaternion-spin quantum cellular automaton) are reframed as **falsifiable bottom-up emergence tests**, each with a protocol and a pre-registered success criterion. Run as toys (E1-E4, all real asserts; counts now 30 + 19 + 3 = 52): **none** reproduces 83/121, 35/437, or Q = 11 without tuned input. The failures are mechanistic - the i-tetration orbit's rotation number is not a low-order rational so the braid does not close (no Jones/FQHE reading); i^z is a 2D map and cannot generate the 3/6 split alone; a bare i-tetration lattice is a pure contraction that collapses to one attractor (no dust/vacuum coexistence). Net: the naive emergence realizations are **ruled out**, and the next target (an explicit entanglement-to-dimension / source rule) is stated precisely. No physics number changed; no fitted parameter added. (**A** = 35/437, **Q** = 11, dim **Z** = 2) LOCKED.

v1.5 (June 2026): The minimal source term for dust/vacuum coexistence. Acts on v1.4's diagnosis (a bare i-tetration contraction erases all dust) by designing and running the *simplest* source rule that lets matter resist the de Sitter collapse (Appendix E.5; CX1-CX4; counts now 30 + 23 + 3 = 56). With the contraction rate fixed by the corpus (gamma = 1 - |**lambda**| = 0.108), a bistable autocatalytic source gives two-phase coexistence *iff* its strength exceeds a corpus-tied threshold **J >= 4(1 - |lambda|) = 0.434** (Maxwell point J_M = (9/2)(1 - |**lambda**|) = 0.488). The minimal rule FORM is thus pinned, but the dust/vacuum *ratio* is still NOT pinned (generically 0 or 1; 83/121 only by tuning) - which *independently reproduces* the §3 No-Go and the G3 gate (the ratio is a present-epoch condition, not a dynamical invariant). Net: a genuine, falsifiable advance on the emergence programme; the next target (derive J from **A**, **Q**) is stated. No physics number changed; no fitted parameter added. (**A** = 35/437, **Q** = 11, dim **Z** = 2) LOCKED.

v1.6 (June 2026): Consolidation + manuscript integration + the rank-weighted master equation. Integrates the v1.5 coexistence result into the body and reframes the paper around its honest central thesis - the **convergence of four independent routes** (two No-Go theorems, the rank-83 flux obstruction, and the reaction-diffusion coexistence analysis) on the conclusion that the budget is a *present-epoch boundary condition*, not a dynamical invariant. **Adds (best G3 candidate):** the rank-weighted master equation (rates into a sector proportional to its rank, $q_{m\text{->}L}$ = 83k, $q_{L\text{->}m}$ = 38k) whose unique, probability-conserving stationary state is (38/121, 83/121) with *no free J* (ME1) - the dynamical form of the §8.2 maximally-mixed candidate rho_* = I/121. It is the structurally correct object (it fixes the ratio where the cubic source cannot), but remains HYPOTHESIS-strong: the equal-amplitude assumption and the occupation-to-energy map are open. **Corrections (prior AI-pass):** E4 (an identity) moved to a note; E1 lowered to “no low-period closure up to the pre-registered bound” (not a no-Jones proof) and the denominator-bound typo fixed; CX4 split into an analytic area-sign assert (CX4a) and a numerical front-velocity note (CX4b); robust bistability stated as J > J_c (J = J_c is a saddle-node threshold); the Maxwell point shown to be unstable to curvature-driven coarsening (so even Maxwell balance does not preserve a seeded fraction); “minimal rule” scoped to “the minimal scalar cubic ansatz.” **Tested and reported negative:** a pre-registered J = f(**A**, **Q**) guard - no independent A,Q expression selects a physical J, and even fixing $J = J_M$ with the ratio seeded does not preserve it, so deriving J is NOT an exit. Counts 30 + 23 + 3 = 56 (+8 notes). No physics number changed; no fitted parameter added. (**A** = 35/437, **Q** = 11, dim **Z** = 2) LOCKED.